Mock Test

NAME (pinyin):________________ NAME (汉字):_______

S&T Batch 2025 Checkpoint 2 Mock Test [100 marks]

1. **<u>Draw</u>** the Born-Harbor cycle of calcium chloride.

   <u>Label</u> the reactant(s) / product(s) with state symbols on the horizontal lines.

   <u>Indicate</u> the **name** of the enthalpies corresponding to each step. For ionization energy and/or electron affinity, indicate its order, e.g. 1^{st}, 2^{nd}, 3^{rd}, …

   <u>Search</u> the numerical value of each enthalpy and **<u>calculate</u>** the lattice energy.

   [12 marks in total]

2. [26 marks] Synthesis of ammonia is crucial to human survival because it is a key component in fertilizers, which are essential for growing the food that sustains the global population.

   a) **<u>Write</u>** the equation for the synthesis of ammonia, with state symbols. [2 marks]

   b) **<u>Calculate</u>** the $\Delta_r H^o$, $\Delta_r S^o$ and $\Delta_r G^o$ of the reaction at T_1 = 673 K. Assume the change of enthalpy and entropy are temperature-independent. [6 marks]

	Nitrogen	Hydrogen	Ammonia
$\Delta_f H^o_{298}$ (kJ mol^{-1})			-45.94
S^o_{298} (J mol^{-1} K^{-1})	191.50	130.57	192.67

c) **<u>Calculate</u>** the equilibrium constant K_1 of the reaction at T_1 = 673 K. [2 marks]

A reaction vessel is kept at constant pressure of 300 bar and constant temperature of T_1 = 673 K. The vessel is initially filled with N_2 and H_2 in the molar ratio of 1:3.

d) **<u>Calculate</u>** the equilibrium partial pressure of each component and hence **<u>determine</u>** the degree of conversion for nitrogen gas. [9 marks]

e) **Calculate** the temperature T_2 (in K) at which one third of nitrogen is converted at equilibrium, given that the total pressure is kept at 300 bar and the same initial molar ratio is utilized. [7 marks]

3. [10 marks] Hess's Law is useful in calculation of enthalpy change of reactions, particularly the reactions that are too dangerous to be experimentally conducted.

   Using Hess's Law, **calculate** $\Delta_r H^o$ of the reaction between liquid methylhydrazine (CH_3-NH-NH_2) and dinitrogen tetraoxide. The products are nitrogen, carbon dioxide and liquid water.

   You may choose to use the following data:

Chemical	N_2O_4 (g)	CO_2 (g)	CO (g)	H_2O (l)	H_2O (g)
$\Delta_f H^o$ (kJ mol^{-1})	9.08	–393.52	–110.53	–285.83	–241.82

Enthalpy of combustion of methylhydrazine (l): –1305.2 kJ mol^{-1} (298 K)

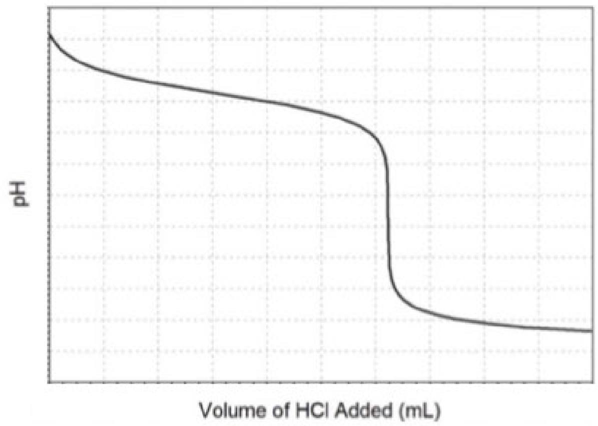
Enthalpy of combustion of carbon monoxide: –283.0 kJ mol^{-1} (298 K)

4. [12 marks] pH control is important not only in chemistry, but also in biology, material science, environmental science and engineering.

   a) **Calculate** the pH of 0.100 mol dm^{-3} HCl solution [1 mark]

b) **<u>Calculate</u>** the pH of 0.100 mol dm^{-3} NH_3 solution (pK_b = 4.75). [2 marks]

10.0 cm^3 of solution b) is titrated with solution a). The pH value is continuously measured and the titration curve is plotted.



c) What is the volume (V_1) of solution a), when end point is reached? [1 mark]

d) Calculate the pH of the end point. [3 marks]

e) Calculate the pH of the mixture, when the volume of the titrant added $V_2 = \frac{2}{3}V_1$. [5 marks]

5. [17 marks] The molecular formula of compound **A** is C_3H_5OCl.

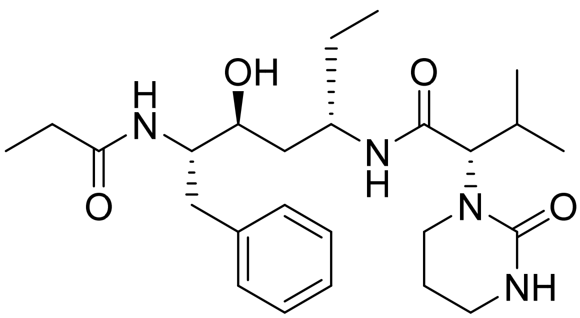
a) **<u>Draw</u>** all acyclic isomers of **A**, which has C=O bond. Do NOT consider stereoisomers. [4 marks]

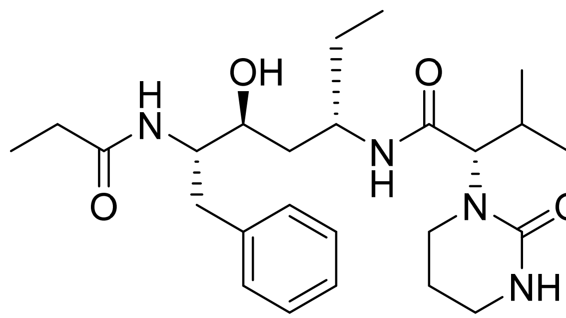
b) **<u>Draw</u>** all stable acyclic isomers of **A**, which has C=C bond. NOT consider stereoisomers. [7 marks]
Hint: Alkenes with hydroxyl group directly attached to alkene sp^2 carbon is unstable. In addition, compounds with hydroxyl group and halogen attached to the same carbon is also unstable.

c) From all the isomers in part a) and b), **<u>draw</u>** the (R) and (Z) isomers below. **<u>Write</u>** the IUPAC name of the (R) isomer(s). [6 marks]

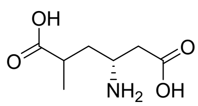
6. [8 marks]
   a) **<u>Write</u>** the molecular formula of the following compound. [2 marks]

   b) For each chiral centre in following molecule, **<u>rank</u>** the groups by their priority and **<u>assign</u>** the absolute configuration. If you need to label multiple chiral centers in one graph, choose two chiral centers far away and use two different colors. [6 marks]

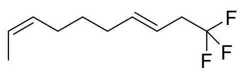




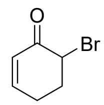
7. [15 marks] **<u>Write</u>** the IUPAC name of the following compounds, paying special attention to stereochemistry when applicable.



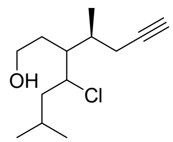
A (3 marks)



B (3 marks)



C (3 marks)



D (3 marks)

E: $(CH_3CH_2CH_2COO)_3Al$

F: Cu_2O

G: H_3PO_4

E,**F**,**G**: 1 mark each

A	
B	
C	
D	
E	
F	
G	

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